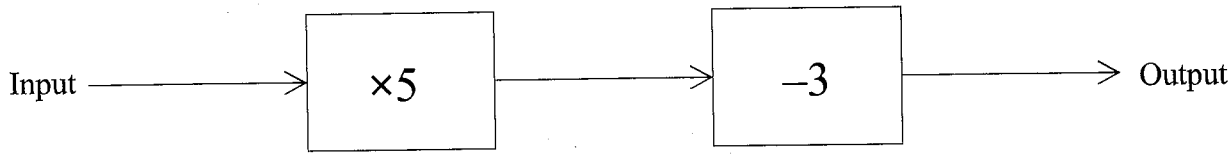
1 Here is a number machine.



(a) What is the **output** when the **input** is 7?

$7 \times 5 = 35$

$35 - 3 = 32$

32 .....

(1)

Here is a different number machine.

When the input is 9 the output is 2.

$9 + 11 = 20$

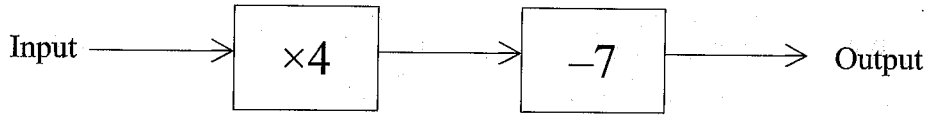
(b) Complete the number machine.

÷ 10 or × 0.1 or − 18

(1)

(Total for question 1 is 2 marks)

2 Here is a number machine.



(a) What is the **output** when the **input** is 6?

6 × 4 = 24
24 – 7 = 17

17
(1)

(b) What is the **input** when the **output** is 25?

Go backwards

25 + 7 = 32
32 ÷ 4 = 8

8
(2)

(Total for question 2 is 3 marks)

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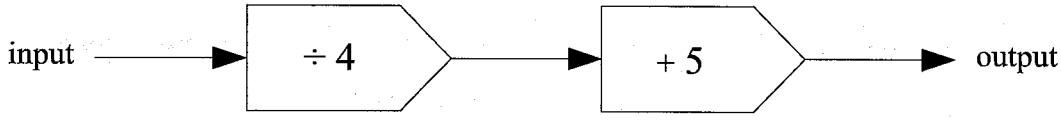
3 Here is a number machine.



(a) Find the **output** when the **input** is 12

12 ÷ 4 = 3
3 + 5 = 8

8
(1)

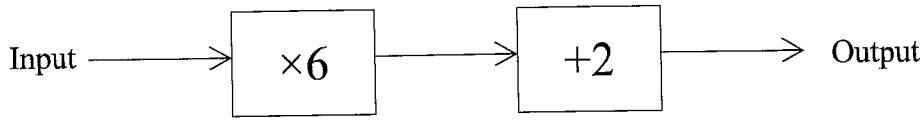
(b) Find the **input** when the **output** is 13

13 – 5 = 8
8 × 4 = 32

32
(2)

(Total for question 3 is 3 marks)

4 Here is a number machine.



(a) What is the **output** when the **input** is 3?

3 × 6 = 18

18 + 2 = 20

20

(1)

(b) What is the **input** when the **output** is 44?

44 − 2 = 42

42 ÷ 6 = 7

7

(2)

(Total for question 4 is 3 marks)

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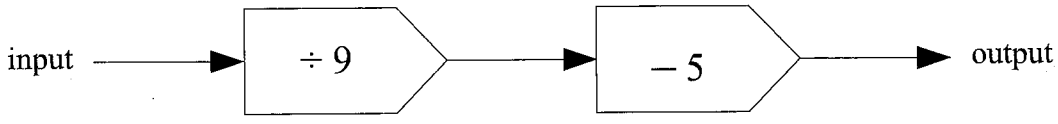
5 Here is a number machine.



(a) Find the **output** when the **input** is 81

81 ÷ 9 = 9

9 − 5 = 4

4

(1)

(b) Find the **input** when the **output** is 0

0 + 5 = 5

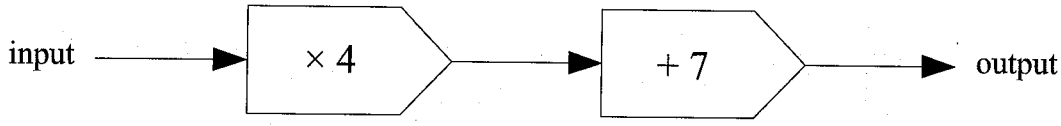
5 × 9 = 45

45

(2)

(Total for question 5 is 3 marks)

6 Here is a number machine.



(a) Find the output when the input is 5

$5 \times 4 = 20$

$20 + 7 = 27$

27

(1)

(b) Find the output when the input is –3

$-3 \times 4 = -12$

$-12 + 7 = -5$

-5

(1)

(c) Find the input when the output is 71

$71 - 7 = 64$

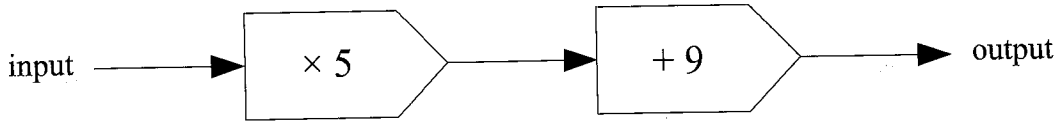
$64 \div 4 = 16$

16

(2)

(Total for question 6 is 4 marks)

7 Here is a number machine.



(a) Find the output when the input is 5

5 × 5 = 25

25 + 9 = 34

34 .....
(1)

(b) Find the output when the input is –2

−2 × 5 = −10

−10 + 9 = −1

−1 .....
(1)

(c) Find the input when the output is 64

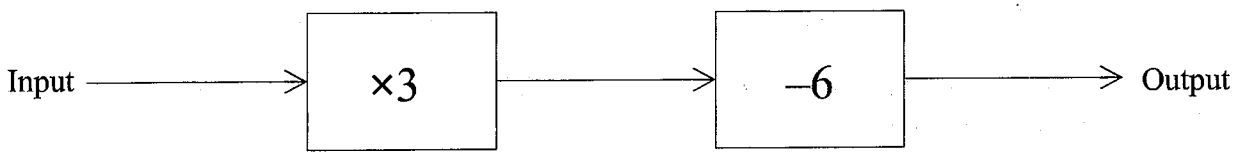
64 − 9 = 55

55 ÷ 5 = 11

11 .....
(2)

(Total for question 7 is 4 marks)

8 Here is a number machine.



(a) What is the **output** when the **input** is 4?

$4 \times 3 = 12$

$12 - 6 = 6$

6 .....
(1)

(b) What is the **input** when the **output** is 15?

$15 + 6 = 21$

$21 \div 3 = 7$

7 .....
(1)

(b) Show that there is a value of the input for which the input and the output have the same value.

Input	Output
2	0
3	3

$3 \times 3 = 9$

$9 - 6 = 3$

$3 \rightarrow \boxed{\times 3} \rightarrow \boxed{-6} \rightarrow 3$

(2)

(Total for question 8 is 4 marks)

9 A rule to change from temperate measured in degrees Celsius (°C) to degrees Fahrenheit (°F) is

Multiply the temperature in degrees Celsius by 1.8 then add 32

input → × 1.8 → + 32 → output

The temperature in London is 12°C.

(a) Work out the temperature, in London, in Fahrenheit (°F)

$12 \times 1.8 = 21.6$

$21.6 + 32 = 53.6$

53.6

(2)

The temperature in New York is 54°F

(b) Work out the temperature, in New York, in Celsius (°C)

$54 - 32 = 22$

$22 \div 1.8 = 12.\dot{2}$

12.2

(2)

(Total for question 9 is 4 marks)

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10 A rule to calculate a taxi fare is

£2.50 plus £2.20 per mile

Input → × 2.20 → + 2.50 → output

(a) Work how much a 10 mile taxi journey would cost.

$10 \times 2.2 = 22$

$22 + 2.50 = 24.50$

£24.50

(1)

A taxi journey costs £20.10

(b) Work out distance of the journey.

$20.10 - 2.50 = 17.60$

$17.60 \div 2.20 = 8$

8

(2)

(Total for question 10 is 3 marks)